

Full-Band Speech Enhancement via Low-Band-Conditioned Refinement

December 2026

Department of Artificial Intelligence
Graduate School, Sogang University
Woo Seok Kim

Full-Band Speech Enhancement via Low-Band-Conditioned Refinement

Advisor : Prof. Hyung-Min Park

A thesis submitted in partial fulfillment of the
requirements for the degree of Master in
Artificial Intelligence

December 2026

Department of Artificial Intelligence

Graduate School, Sogang University

Woo Seok Kim

논 문 인 준 서

김우석의 공학석사 학위논문을 인준함

2026 년 12 월

주심 김 지 환 ㉠

부심 장 두 성 ㉠

부심 박 형 민 ㉠

Contents

Contents	i
List of Figures	iii
List of Tables	iv
Abstract	v
1 Introduction	1
2 Related Work	5
2.1 Full-Band Speech Enhancement	5
2.2 Efficient Full-Band Modeling	6
2.3 Query Refinement and TF Filtering	7
3 Proposed Method	9
3.1 Problem Formulation	9
3.2 TF-Rehancer Architecture	10
3.2.1 SFI-STFT Conditioning and Shared Embedding	10
3.2.2 Observed High-Band Query Construction	13
3.2.3 Cross-Band Decoder Refinement	16

3.2.4	Complex TF Filtering	18
3.3	Training Objective	19
4	Experiments	22
4.1	Experimental Setup	22
4.1.1	Datasets and Protocols	22
4.1.2	Model Configuration	23
4.1.3	Evaluation Metrics	24
4.2	Experimental Results	25
4.2.1	48 kHz Full-Band Results	25
4.2.2	Wideband Results	26
4.2.3	Low-Band Cutoff	27
4.2.4	Architecture Ablations	27
5	Conclusion	29
5.1	Summary	29
5.2	Limitations and Future Work	30
	Bibliography	32

List of Figures

1.1	Conceptual overview of TF-Rehancer	3
3.1	Overall TF-Rehancer architecture	11
3.2	Local convolutional layer notation	13
3.3	TF-Rehancer encoder and decoder core blocks	16
3.4	Complex TF filtering module	18

List of Tables

4.1	48 kHz full-band results on VoiceBank+DEMAND 48 kHz	25
4.2	VoiceBank+DEMAND 16 kHz test results	26
4.3	Low-band cutoff ablation	27
4.4	48 kHz architecture ablation	28

Abstract

This thesis presents TF-Rehancer, a time-frequency speech enhancement architecture for direct full-band processing. The model treats high-frequency components in 48 kHz speech as observed but noisy spectral evidence rather than as missing information to be generated from a band-limited input. TF-Rehancer performs low-band-guided full-band refinement: it encodes low-band speech structure, preserves observed upper-band evidence in the decoder path, and estimates the enhanced spectrum through local complex time-frequency filtering over the original noisy STFT.

We evaluate TF-Rehancer from three perspectives. First, on the 16 kHz VoiceBank+DEMAND benchmark, the compact FastGRU-based TF-Rehancer configuration achieves PESQ 3.33, DNSMOS SIG 3.41, DNSMOS BAK 4.04, and DNSMOS OVL 3.13 with 0.47M parameters and 1.89G model-side MACs. Against the paper-reported FastEnhancer-M reference, it reports higher PESQ, SI-SDR, and DNSMOS P.835 values with fewer model-side MACs, while FastEnhancer-M remains slightly stronger in SCOREQ and ESTOI.

Second, on the VoiceBank+DEMAND 48 kHz test set, the DNS4-trained online TF-Rehancer configuration obtains competitive full-reference scores relative to published PercepNet and DeepFilterNet2 references. Third, controlled 48 kHz ablations show that replacing complex TF filtering with direct spectrum mapping degrades full-reference, spectral, and non-intrusive metrics. They also show that using decoder cross-attention is more effective than removing it on most reported full-reference and spectral metrics, and that a 5 kHz low-band cutoff provides a balanced operating point under the controlled ablation setting.

Chapter 1

Introduction

Many established speech enhancement benchmarks and baselines are built around conventional 16 kHz wideband settings. In this setting, the upper-band spectral components contained in 48 kHz full-band speech are outside the model input. Direct 48 kHz full-band speech enhancement therefore presents a different operating condition: the model receives a noisy full-band signal, and the upper-band spectrum is available as noisy evidence to be enhanced. Although full-band enhancement has been explored in prior work, same-protocol references for direct 48 kHz full-band SE remain less established than conventional wideband evaluation.

Speech bandwidth extension and speech super-resolution target a different information condition. These tasks aim to reconstruct missing or degraded high-frequency content from band-limited inputs [1, 2], whereas 48 kHz full-band SE starts from a noisy full-band observation. Therefore, the key issue is not to synthesize the upper band after wideband enhancement, but to suppress noise over the full observed spectrum while preserving useful

spectral evidence. This distinction motivates a direct full-band enhancement approach in which the high-frequency region is treated as part of the noisy observation rather than as a missing component to be generated afterward.

Practical full-band systems such as PercepNet and DeepFilterNet2 have shown that full-band enhancement can be made feasible under strict computational constraints [3, 4]. PercepNet uses perceptually motivated critical-band features and comb filtering, while DeepFilterNet2 combines ERB-domain enhancement with deep filtering. Such perceptual-band front ends trade bin-level spectral resolution for efficiency, which motivates a complementary TF-domain design that uses a learned frequency representation and estimates the enhanced spectrum through complex filtering over the original noisy STFT.

Directly applying heavy TF modeling to all full-band frequency tokens can be computationally costly, especially as the number of STFT bins increases with sampling rate. Since harmonic and formant-related speech-structure evidence is concentrated mainly in lower frequencies while upper-frequency bins remain observed in the noisy input, a natural design is to concentrate heavier encoding on low-band structure and carry the high band as decoder-side observed-query evidence.

In this work, we propose TF-Rehancer, a refinement-oriented time-frequency architecture for direct full-band speech enhancement. The core idea is low-band-guided full-band refinement. TF-Rehancer concentrates heavier encoding on low-band speech structure, preserves observed upper-band evidence in the decoder path, and estimates the enhanced spectrum through complex filtering over the original noisy STFT. This design allows the model to exploit cross-band speech structure while keeping the final estimation tied to the observed noisy signal.

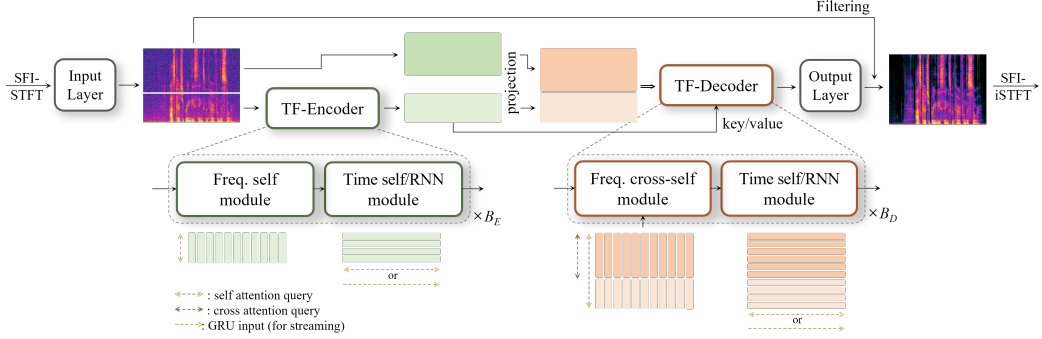


Figure 1.1: Conceptual overview of TF-Rehancer. The low-band encoder output provides key/value context for updating the observed high-band decoder tokens. The updated representation is then processed by full-sequence decoder modules and used to estimate local complex filters over the original noisy STFT.

From the attention perspective, the low-band encoder output provides key/value context for the high-band portion of the decoder representation. Cross-attention updates the observed high-band decoder tokens, after which full-sequence decoder modules model the joint low- and high-band representation. Thus, TF-Rehancer does not generate high-band speech from low-band information alone; it uses low-band context to condition the update of observed high-band features.

We evaluate TF-Rehancer in three settings. The first is 48 kHz full-band results on the VoiceBank+DEMAND 48 kHz test set against published full-band references. The second is a 16 kHz VoiceBank+DEMAND quality-efficiency comparison. The third is controlled 48 kHz ablation analysis of the proposed architecture.

The contributions of this work are summarized as follows.

1. We formulate 48 kHz full-band SE as observation-preserving refinement, where upper-band components are treated as noisy spectral evidence rather than missing content to be generated after wideband processing.

2. We propose TF-Rehancer, a TF-domain architecture for low-band-guided full-band refinement. It updates observed high-band decoder tokens using low-band encoder context, models the resulting full embedded sequence, and estimates the enhanced spectrum through complex filtering over the noisy STFT.
3. We evaluate TF-Rehancer through 48 kHz full-band results against published full-band references, a 16 kHz quality-efficiency comparison, and controlled 48 kHz ablations of complex filtering, decoder cross-attention, and the low-band cutoff.

The remainder of this thesis is organized as follows. Chapter 2 reviews related work in wideband enhancement, bandwidth extension, and practical full-band enhancement. Chapter 3 describes the TF-Rehancer architecture and training objective. Chapter 4 presents the experimental setup, benchmark results, and ablation analysis. Chapter 5 summarizes the thesis and discusses limitations and future work.

Chapter 2

Related Work

2.1 Full-Band Speech Enhancement

Neural speech enhancement (SE) has evolved from time-frequency mask estimation to phase-aware, complex-domain, time-domain, and modern TF-domain modeling. Early systems commonly estimated real-valued masks in the STFT domain, while later methods introduced phase-sensitive masks, complex ratio masks, and complex-valued networks to better handle phase-related distortion. Time-domain systems replaced the fixed STFT representation with learned analysis and synthesis bases, and recent TF-domain models further improved enhancement quality by explicitly modeling temporal and spectral dependencies [5, 6, 7, 8, 9, 10, 11]. However, many reported comparisons remain tied to 16 kHz or protocol-specific settings, where the upper-band components present in 48 kHz speech are outside the input representation.

The motivation for full-band SE is also supported by perceptual studies on high-

frequency speech information. Extended high-frequency cues above the conventional wide-band range have been shown to contribute to speech perception in noise and to the perception of fricative-related cues [12, 13, 14]. These findings motivate retaining and refining upper-band observations when the full-band spectrum is already present in the input.

Speech bandwidth extension and speech super-resolution address a different information condition from direct full-band SE. These tasks aim to reconstruct or supplement missing high-frequency content from band-limited inputs, often targeting a higher output sampling rate or wider output bandwidth [1, 2, 15]. In contrast, 48 kHz full-band SE starts from a noisy full-band observation, where the upper-band spectrum is already present but corrupted by noise. Therefore, full-band SE requires noise suppression and spectral refinement over the observed full-band signal, rather than only high-band reconstruction from a band-limited input. This observed-band condition motivates architectures that retain noisy full-band evidence while using reliable low-band speech structure as refinement context.

2.2 Efficient Full-Band Modeling

Practical full-band SE has been studied mainly under real-time and low-complexity constraints. PercepNet enhances 48 kHz speech using perceptually motivated critical-band features and comb filtering, while DeepFilterNet2 combines ERB-domain enhancement with deep filtering for low-complexity full-band enhancement [3, 4]. Such perceptual or ERB-band representations trade frequency resolution for efficiency, motivating complementary TF-domain designs that retain a closer connection to the observed STFT while still controlling computational cost.

Frequency-structured modeling has also become a major design axis in modern SE and source separation. Subband and band-split models process spectra through grouped frequency regions, while TF-domain dual-path models separate temporal and spectral modeling to improve efficiency and representation power. BSRNN and BS-RoFormer show that band-wise modeling is effective for high-resolution audio modeling, and TF-Locoformer develops an RNN-free TF-domain dual-path architecture with local modeling by convolution [16, 17, 11]. Lightweight systems such as GTCRN, LiSenNet, and FastEnhancer further show that compact recurrent, subband, and frequency-time modeling can improve the performance-efficiency trade-off in real-time SE [18, 19, 20].

Many band-split and lightweight models emphasize compact subband grouping or efficient within-band/global sequence modeling. TF-Rehancer also uses frequency compression: its input stem applies a learned frequency-strided embedding to reduce token count. The difference is that the final enhancement remains tied to the original noisy STFT through full-resolution complex filtering. This design targets direct 48 kHz full-band SE, where the high-frequency region is not missing but available as noisy spectral evidence in the decoder path.

2.3 Query Refinement and TF Filtering

Query-based restoration provides another relevant direction. TF-Restormer introduces an asymmetric encoder-decoder architecture for speech restoration with decoupled input and output sampling rates. Its encoder focuses on the observed input bandwidth, while the decoder restores target frequency regions through frequency extension queries [21]. This mechanism is well matched to bandwidth restoration, where decoder-side queries

can serve as slots for frequency regions that are absent from the input.

Direct 48 kHz full-band SE has a different information condition. TF-Restormer uses decoder-side queries for missing or target frequency regions in restoration, whereas TF-Rehancer uses query modeling in direct full-band SE, where the high-band region is observed and noisy. The query is therefore not a missing-band placeholder; it is derived from observed high-band evidence and updated using low-band context.

Filtering-based enhancement is closely related to this observation-preserving view. Instead of directly mapping a latent representation to a clean spectrum, deep filtering estimates complex local filters and applies them to the noisy STFT [4]. Recent correlation-to-filter approaches such as SR-CorrNet provide related evidence that correlation-derived or structure-conditioned filters can be effective alternatives to direct spectral mapping [22]. TF-Rehancer follows this filtering-based principle by estimating the enhanced spectrum through complex filtering over the original noisy STFT.

Prior work covers complementary parts of this problem. Practical full-band systems emphasize perceptual-band efficiency, frequency-structured models mainly address efficient within-band or band-split modeling, and query-based restoration methods mainly target missing-band reconstruction. In contrast, TF-Rehancer targets direct full-band SE through low-band-guided full-band refinement and complex filtering over the noisy STFT.

Chapter 3

Proposed Method

3.1 Problem Formulation

Let $y \in \mathbb{R}^L$ and $x \in \mathbb{R}^L$ denote the noisy and clean speech waveforms, respectively. In single-channel speech enhancement, the noisy signal is modeled as

$$y = x + n, \tag{3.1}$$

where n denotes additive noise. Applying the short-time Fourier transform (STFT) gives noisy and clean complex spectra $Y, X \in \mathbb{C}^{F \times T}$, where F and T denote frequency bins and time frames. The target of full-band speech enhancement is to estimate a clean full-band spectrum $\hat{X}$ from the noisy full-band observation Y , followed by inverse STFT reconstruction of the enhanced waveform $\hat{x}$.

TF-Rehancer follows an observation-preserving filtering formulation. The neural network predicts a complex local time-frequency filter W , and the final spectrum is obtained

by applying this filter to a linear noisy STFT reference Y^{ref} :

$$W = \mathcal{G}_\theta(\Phi(Y)), \quad \hat{X} = W \otimes Y^{\text{ref}}. \quad (3.2)$$

Here, $\mathcal{G}_\theta$ denotes TF-Rehancer, $\Phi(Y)$ is the neural conditioning feature, and $\otimes$ denotes local complex filtering over a time-frequency neighborhood. The compressed feature $\Phi(Y)$ is used for conditioning, while the output operator is applied to the original linear noisy spectrum. Therefore, the model is not trained as an unconstrained direct generator of $\hat{X}$; it estimates the filter coefficients used to transform the observed noisy spectrum.

For a 48 kHz full-band input, the upper band is already present in Y , although it is corrupted by noise. TF-Rehancer treats this region as observed noisy evidence, derives decoder-side high-band query features from it, and refines the decoder representation before complex filtering.

3.2 TF-Rehancer Architecture

This section describes TF-Rehancer as a low-band-guided full-band refinement pipeline: shared TF embedding, observed high-band query construction, cross-band decoder refinement, and full-resolution complex TF filtering.

3.2.1 SFI-STFT Conditioning and Shared Embedding

TF-Rehancer adopts the sampling-frequency-independent STFT (SFI-STFT) principle used in TF-Restormer [21]. The analysis window and hop are fixed by physical duration, 40 ms and 20 ms, rather than by a fixed sample count. Thus, the 16 kHz configuration

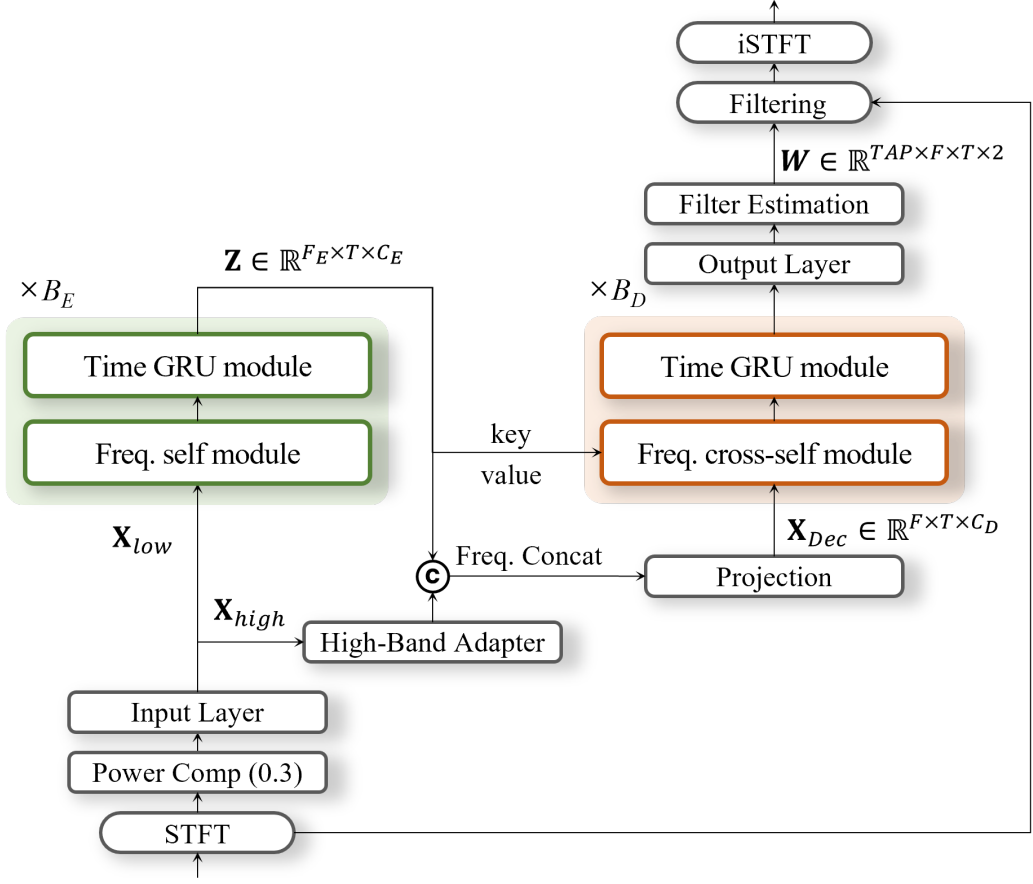


Figure 3.1: Overall TF-Rehancer architecture. The shared input layer embeds the full-band STFT. The low-band region is encoded by the TF encoder, while the observed high-band region is carried as decoder-side query evidence. Decoder cross-attention updates the high-band tokens using low-band encoder context, and subsequent full-sequence decoder modeling produces features for complex TF filter estimation.

uses $N = 640$, $H = 320$, and $F = 321$, while the 48 kHz configuration uses $N = 1920$, $H = 960$, and $F = 961$. This keeps the temporal frame geometry fixed across sampling rates and lets the frequency-bin count scale with bandwidth. Unlike TF-Restormer, which uses decoupled input-output rates for restoration and super-resolution, TF-Rehancer uses the same SFI-STFT parameterization for the noisy input, clean target, and inverse-STFT

reconstruction in same-sampling-rate 16 kHz and 48 kHz speech enhancement settings.

Figure 3.1 summarizes the TF-Rehancer architecture used in this thesis. The frontend constructs a power-law compressed conditioning feature while preserving a linear noisy STFT reference for the final filtering step. The shared input layer embeds the full-band STFT and reduces only the frequency resolution. A fixed 5 kHz split then routes low-band tokens to the TF encoder and observed high-band tokens to the decoder-side query path. The encoder uses four low-band time-frequency blocks to model speech-structure evidence. The decoder uses two cross-band time-frequency blocks that first update only the high-band decoder tokens using low-band key/value context and then model the full embedded sequence. Finally, FastFilterUpsample restores the original STFT frequency resolution, and the output layer estimates a local 3×3 complex filter applied to Y^{ref} .

The neural conditioning feature is constructed from a power-law compressed complex STFT. Given $\gamma = 0.3$, the compressed spectrum and input feature are

$$C_\gamma(Y) = Y (|Y|^2 + \epsilon)^{(\gamma-1)/2}, \quad \Phi(Y) = [\text{Re}(C_\gamma(Y)), \text{Im}(C_\gamma(Y)), |C_\gamma(Y)|]. \quad (3.3)$$

Power-law compression stabilizes the dynamic range of spectral amplitudes for neural processing. The uncompressed noisy STFT reference Y^{ref} is kept separately for the final filtering operation.

The feature $\Phi(Y)$ is passed through a shared frequency-strided input stem, denoted by $H = \text{Stem}(\Phi(Y))$. The stem consists of Conv2D($3 \rightarrow 48$), batch normalization, and a SiLU activation. The convolution uses kernel size $(4, 1)$, stride $(2, 1)$, and padding $(2, 0)$, so it reduces only the frequency resolution and preserves the frame rate. This shared stem is used for both 16 kHz and 48 kHz configurations. After the stem, the embedded frequency

resolution is $F_{\text{emb}} = 161$ at 16 kHz and $F_{\text{emb}} = 481$ at 48 kHz. The shared stem compresses the frequency axis for efficient modeling while deriving high-band decoder tokens from the observed high-band portion of the input representation.

On the output side, the decoder representation is restored to the original STFT frequency resolution before estimating local complex filter coefficients. Thus, both input-side embedding and output-side filtering are described as local convolutional transformations around an explicit STFT reference, rather than as unconstrained waveform generation.

3.2.2 Observed High-Band Query Construction

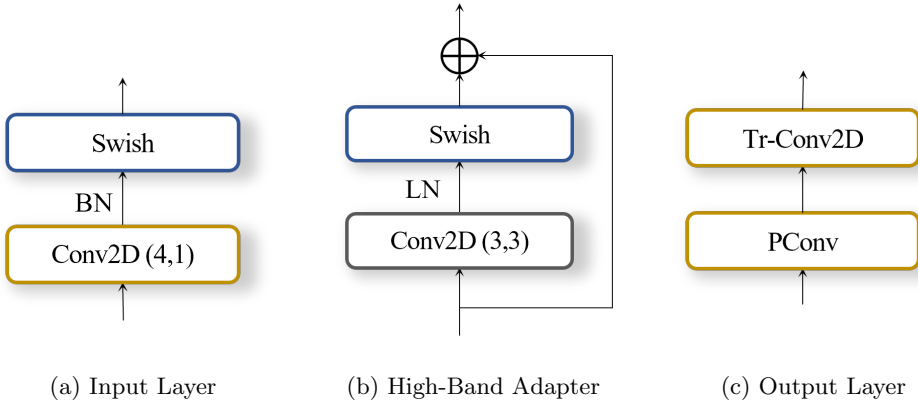


Figure 3.2: Local convolutional layer notation used for the input stem, high-band adapter, and output-side upsampling/filtering layers. The channel width, stride, and normalization differ by module and are specified in the text.

After the shared stem, TF-Rehancer splits the embedded frequency axis using a 5 kHz cutoff into H_{low} and H_{high} . At 16 kHz, this gives $F_{\text{low}} = 101$ and $F_{\text{high}} = 60$. At 48 kHz, it gives $F_{\text{low}} = 101$ and $F_{\text{high}} = 380$. Thus, the same low-band speech-structure region is encoded at both sampling rates, while the 48 kHz model keeps a much larger observed high-band region in the decoder path.

The high-band branch is derived from the observed high-band stem feature:

$$Q_h = H_{\text{high}} + \text{Adapter}_h(H_{\text{high}}). \quad (3.4)$$

The adapter is a residual local convolutional block with SiLU activation and layer normalization, using the local convolutional layer notation summarized in Fig. 3.2. This block aligns the observed high-band representation with the decoder query space before cross-band refinement. It does not perform decoder refinement by itself; it prepares the high-band tokens for the decoder. The query source remains the observed high-band embedding, and the adapter only provides a lightweight local transformation before cross-band refinement.

We next describe the low-band encoder that produces the context representation used by the decoder cross-attention. The low-band encoder maps H_{low} to an encoded representation Z using four stacked blocks. Each block combines frequency feed-forward layers, frequency-axis self-attention with RoPE, and FastGRU2 temporal modeling. The encoder does not use a frequency projection layer or a pre-encoder sinusoidal positional encoding. Its role is to model speech structure from the low-band region, where harmonic and formant-related evidence is relatively dense.

The encoder frequency module follows a Macaron-style structure: a convolutional feed-forward layer is placed before and after frequency-axis self-attention. This F-ConvFFN–F-MHSA–F-ConvFFN ordering lets local convolutional mixing and global frequency attention complement each other. The decoder frequency module extends this pattern by inserting cross-attention before full-sequence self-attention, i.e., F-MHCA–F-MHSA–F-ConvFFN.

RoPE is applied to frequency-axis attention to encode the relative order of frequency tokens without relying on a learned absolute positional table. This is useful because the frequency axis has a fixed physical ordering, while the number of tokens differs between 16 kHz and 48 kHz settings. In decoder cross-attention, the high-band query and low-band key/value tokens occupy different frequency regions, so RoPE offsets preserve their relative frequency positions during cross-band attention.

To reduce the cost of the frequency feed-forward layers, TF-Rehancer uses an efficient convolutional FFN. Instead of applying a large convolutional FFN over the full frequency sequence, the module first downsamples the frequency-axis sequence by adaptive average pooling. It then applies grouped gated 1-D convolutions on the shortened sequence, using separate value and gate branches followed by SiLU gating. We use group convolution with two groups to reduce computation. The output is projected back, upsampled to the original frequency length, and added as a scaled residual correction. This design keeps a full-length residual path while applying the expensive gated convolutional correction on a shorter sequence.

3.2.3 Cross-Band Decoder Refinement

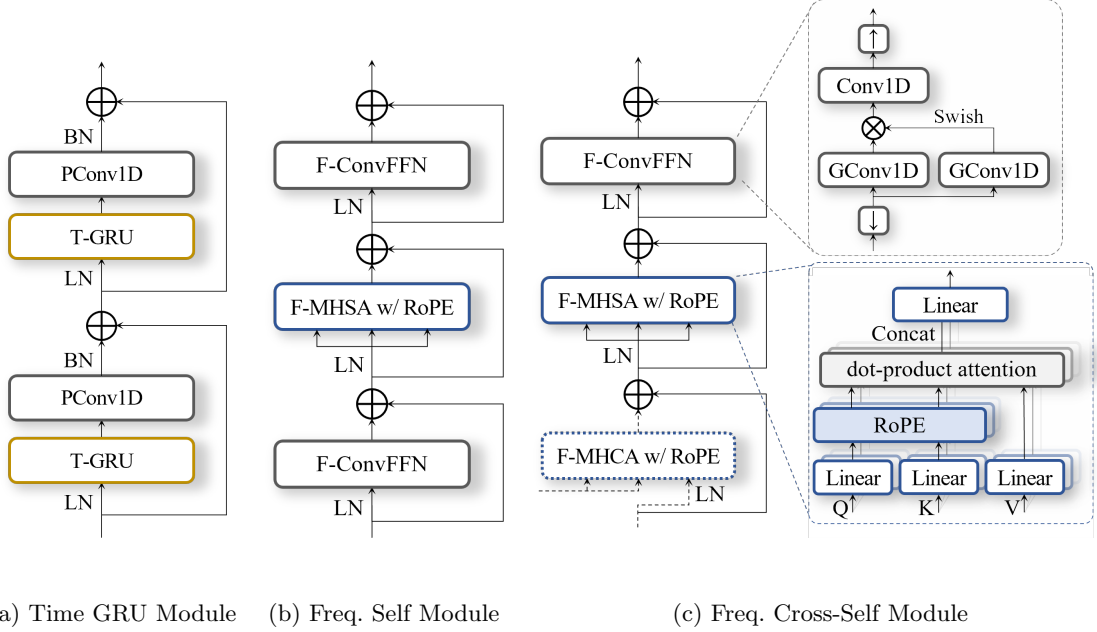


Figure 3.3: TF-Rehancer encoder and decoder core blocks. The encoder frequency module uses a Macaron-style F-ConvFFN–F-MHSA–F-ConvFFN structure with RoPE. The decoder frequency module first applies F-MHCA to update high-band tokens using low-band encoder context, then applies full-sequence F-MHSA and F-ConvFFN.

The decoder first forms a full embedded sequence by concatenating the low-band encoder output Z and the adapted high-band query Q_h along the frequency axis. A linear projection with layer normalization maps the concatenated 48-channel representation to the 24-channel decoder width. This projection is a separate alignment stage between the encoder/query interface and the decoder blocks.

Let the projected decoder input be split as $D = [D_{\text{low}}; D_{\text{high}}]$, where D_{high} corresponds to the observed high-band portion. TF-Rehancer uses two decoder blocks. In each block, D_{low} passes through the cross-attention stage without being updated by MHCA, while D_{high} is used as the query. The key and value are obtained from the low-band encoder

output Z :

$$Q = \Pi_q(D_{\text{high}}), \quad K = \Pi_k(Z), \quad V = \Pi_v(Z). \quad (3.5)$$

The high-band update is then computed as

$$\tilde{D}_{\text{high}} = D_{\text{high}} + \text{MHCA}(Q, K, V), \quad (3.6)$$

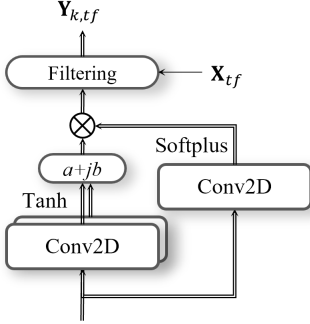
and the decoder sequence is reconstructed as

$$\tilde{D} = [D_{\text{low}}; \tilde{D}_{\text{high}}]. \quad (3.7)$$

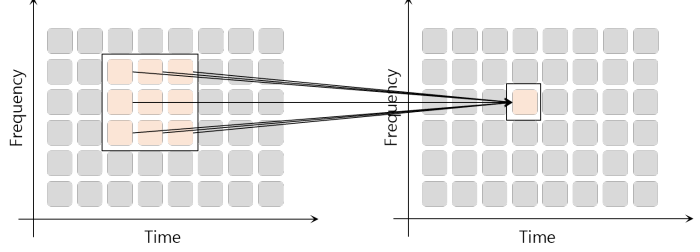
The projection operators Π_q , Π_k , and Π_v are implemented inside the MHCA module, so the input widths of the decoder-side query and low-band context need not be identical. Cross-attention therefore performs directed low-to-high conditioning by updating only the observed high-band decoder tokens. The following frequency self-attention, frequency feed-forward processing, and FastGRU2 temporal modeling operate on $\tilde{D}$, so low- and high-band tokens interact over the full embedded sequence after the directed update.

This design separates three roles. The high-band adapter prepares observed high-band query evidence from H_{high} . The concat-projection layer aligns the low-band encoder output and high-band query into the decoder width. The decoder blocks then perform cross-attention-based high-band token update followed by full-sequence time-frequency modeling.

3.2.4 Complex TF Filtering



(a) Filtering Module



(b) Local 3×3 Complex TF Filtering

Figure 3.4: Complex TF filtering module. TF-Rehancer predicts complex filter coefficients and applies them to a local time-frequency neighborhood of the noisy STFT reference.

The decoder output is still at embedded frequency resolution. Before filter estimation, TF-Rehancer restores it to the original STFT frequency resolution using FastFilterUp-sample. This module uses a pointwise convolution followed by transposed convolution along the frequency axis. The restored full-resolution feature is then passed to a partial-convolution filter head and a local 3×3 time-frequency convolution. The output-side convolutional operations follow the compact local-layer notation in Fig. 3.2, while the explicit filtering operation is described below.

For each time-frequency bin, the head predicts nine complex taps over the local offsets $\Delta f, \Delta t \in \{-1, 0, 1\}$. The raw head output has 27 channels: one magnitude scale, one real component, and one imaginary component for each tap. A softplus activation constrains the magnitude scale to be positive, and tanh activations bound the real and imaginary components before forming the complex coefficients $W_{\Delta f, \Delta t, f, t}$. The enhanced spectrum

is computed as

$$\hat{X}_{f,t} = \sum_{\Delta f=-1}^1 \sum_{\Delta t=-1}^1 W_{\Delta f, \Delta t, f, t} Y_{f+\Delta f, t+\Delta t}^{\text{ref}}. \quad (3.8)$$

Boundary positions are padded before local neighborhoods are gathered. This operation is a complex filtering operation, not a correlation estimate. The coefficients W are the predicted filter weights used to construct $\hat{X}$; they are conditioned on the decoder feature and applied to the observed noisy STFT reference.

The filtering head is therefore the output estimator of the proposed refinement architecture, while the main representation-level contribution lies in low-band-guided full-band refinement.

3.3 Training Objective

TF-Rehancer is trained with a composite enhancement loss using the same $\gamma = 0.3$ power-law compression as the input feature, with an ϵ -stabilized magnitude. Let $\hat{X}$ and X denote the estimated and clean complex STFTs, and let $\hat{x} = \text{iSTFT}(\hat{X})$ and $x = \text{iSTFT}(X)$ denote the corresponding waveforms. The total enhancement loss is

$$\mathcal{L} = \lambda_{\text{mag}} \mathcal{L}_{\text{mag}} + \lambda_{\text{cplx}} \mathcal{L}_{\text{cplx}} + \lambda_{\text{cons}} \mathcal{L}_{\text{cons}} + \lambda_{\text{wav}} \mathcal{L}_{\text{wav}} + \lambda_{\text{pesq}} \mathcal{L}_{\text{pesq}}. \quad (3.9)$$

Using $C_\gamma(\cdot)$ for the compressed complex spectrum and $\text{RI}(\cdot)$ for real-imaginary channel stacking, the spectral terms are

$$\mathcal{L}_{\text{mag}} = \text{MSE} \left(|C_\gamma(\hat{X})|, |C_\gamma(X)| \right), \quad (3.10)$$

$$\mathcal{L}_{\text{cplx}} = \text{MSE} \left(\text{RI}(C_\gamma(\hat{X})), \text{RI}(C_\gamma(X)) \right). \quad (3.11)$$

$\mathcal{L}_{\text{mag}}$ matches the compressed magnitude envelope and encourages energy consistency across frequency bands, including the upper band. It does not directly supervise phase. $\mathcal{L}_{\text{cplx}}$ compares the compressed real and imaginary components, so it gives phase-sensitive supervision for the complex filter output.

The consistency term is computed after waveform reconstruction:

$$\tilde{X} = \text{STFT} \left(\text{iSTFT}(\hat{X}) \right), \quad \mathcal{L}_{\text{cons}} = \text{MSE} \left(\text{RI}(C_\gamma(\tilde{X})), \text{RI}(C_\gamma(X)) \right). \quad (3.12)$$

Thus, $\mathcal{L}_{\text{cons}}$ does not merely compare two pre-reconstruction STFT tensors. It penalizes spectra that appear plausible before inverse STFT but become inconsistent after the waveform is reconstructed and transformed back to the STFT domain.

The waveform and perceptual auxiliary terms are

$$\mathcal{L}_{\text{wav}} = \|\hat{x} - x\|_1, \quad \mathcal{L}_{\text{pesq}} = \text{PesqLoss}(\mathcal{R}_{16}(\hat{x}), \mathcal{R}_{16}(x)), \quad (3.13)$$

where $\mathcal{R}_{16}$ denotes resampling to 16 kHz when the training waveform is 48 kHz and the identity operation for 16 kHz training. $\mathcal{L}_{\text{wav}}$ reduces sample-level reconstruction error and time-domain artifacts that may not be fully captured by spectral losses. $\mathcal{L}_{\text{pesq}}$ is a

differentiable PESQ-like training loss, not the reported evaluation PESQ score, and is used only as a weak perceptual regularizer.

The loss weights for $\mathcal{L}_{\text{mag}}$, $\mathcal{L}_{\text{cplx}}$, $\mathcal{L}_{\text{cons}}$, $\mathcal{L}_{\text{wav}}$, and $\mathcal{L}_{\text{pesq}}$ are 0.3, 0.2, 0.3, 0.2, and 0.001, respectively. Therefore, optimization is driven mainly by compressed spectral reconstruction, reconstruction consistency, and waveform-domain supervision, while the PESQ term remains lightly weighted.

Chapter 4

Experiments

4.1 Experimental Setup

4.1.1 Datasets and Protocols

We evaluate TF-Rehancer using 48 kHz full-band results, a 16 kHz wideband benchmark, and a 48 kHz controlled ablation setting. The 48 kHz results use the VBD 48 kHz test set to compare TF-Rehancer with published full-band references and to analyze architectural components under full-band processing. The 16 kHz setting uses the official VoiceBank+DEMAND (VBD) corpus to compare the proposed model with lightweight speech enhancement baselines.

For the 16 kHz comparison, we use the official VBD paired speech enhancement corpus. The official training partition contains 28 speakers and is split into 10,413 training utterances and 1,159 validation utterances using a fixed random seed. Evaluation is performed on the official 824-utterance test set, which contains unseen speakers.

For the controlled 48 kHz ablation experiments, noisy mixtures are synthesized from VCTK [23] clean speech and DNS [24] full-band noise. Evaluation is performed on the VBD 48 kHz test set. The VBD 48 kHz test set is the official 824-utterance Voice-Bank+DEMAND test split prepared at 48 kHz with paired clean and noisy files, not a DNS-remixed test set. The ablation experiments are intended for controlled component analysis.

4.1.2 Model Configuration

This chapter evaluates the final TF-Rehancer architecture described in Chapter 3. The main design principle is low-band-guided full-band refinement. The model concentrates heavy encoding on the low-band representation, keeps observed high-band evidence in the decoder path, updates the high-band decoder tokens using the low-band encoder output as key/value context, and estimates the enhanced spectrum through complex TF filtering over the original noisy STFT.

Unless otherwise specified, TF-Rehancer refers to the architecture that performs low-band-guided full-band refinement and complex TF filtering. For the 16 kHz comparison, we report a compact FastGRU-based configuration and a non-streaming TimeMHSA-EFN configuration; both variants share the same high-band update and filtering pipeline. These labels describe model configurations and are not used as measured real-time or streaming claims.

All TF-Rehancer experiments use a 40 ms analysis window and a 20 ms frame shift. With this setting, both 16 kHz and 48 kHz configurations have a frequency spacing of 25 Hz. The default low-band cutoff is 5 kHz, corresponding to 201 full-resolution frequency

bins and 101 embedded low-band tokens after the frequency-strided stem. Thus, the low-band encoder length is nearly fixed across 16 kHz and 48 kHz configurations, while the number of observed high-band decoder tokens increases in the 48 kHz setting.

Model complexity is reported using the number of parameters and model-side multiply-accumulate operations (MACs). Unless otherwise stated, MACs exclude STFT and inverse STFT operations.

4.1.3 Evaluation Metrics

For locally trained TF-Rehancer runs, evaluation uses the checkpoint with the lowest validation loss. The default VoiceBank+DEMAND evaluation set contains 824 utterances.

We use both full-reference and non-intrusive metrics. DNSMOS P.808 and DNSMOS P.835 SIG/BAK/OVL are computed after resampling the waveform to 16 kHz. SCOREQ is computed in full-reference natural-speech mode after 16 kHz resampling. PESQ, STOI, and ESTOI are also computed after 16 kHz resampling. Unless otherwise specified, SI-SDR is computed at the task sampling rate.

For the 48 kHz analysis, we additionally report full-band spectral metrics because many standard speech enhancement metrics are computed after 16 kHz resampling and do not directly evaluate the upper-frequency region. We report full-band LSD, MCD, and SI-SDR for 48 kHz outputs. DNSMOS SIG/BAK/OVL refers to DNSMOS P.835, while CSIG/CBAK/COVL refers to composite full-reference metrics.

4.2 Experimental Results

4.2.1 48 kHz Full-Band Results

Table 4.1 reports 48 kHz full-band results on the VBD 48 kHz test set together with PercepNet and DeepFilterNet2 references. The TF-Rehancer (on) row is the DNS4-trained online configuration evaluated on the VBD 48 kHz test set.

Method	Params↓	MACs	PESQ↑	STOI↑	CSIG↑	CBAK↑	COVL↑	SI-SDR↑
PercepNet [3, 4]	8.00M	0.80G	2.73	–	–	–	–	–
DFN2 + Simplified DNN [4]	2.31M	0.36G	3.08	0.943	4.30	3.40	3.70	15.709
DFN2 + Post-Filter [4]	2.31M	0.36G	3.03	0.941	3.72	3.37	3.63	15.769
TF-Rehancer (on)	0.615M	3.15G	3.233	0.951	4.286	3.567	3.771	17.789

Table 4.1: 48 kHz full-band results on the VoiceBank+DEMAND 48 kHz test set. PercepNet and DeepFilterNet2 rows provide published-reference context, with DeepFilterNet2 SI-SDR values taken from the official-checkpoint evaluation summary; TF-Rehancer (on) is locally evaluated. MACs are model-side estimates and exclude STFT/iSTFT.

Table 4.1 shows that TF-Rehancer (on) obtains competitive full-reference scores on the VBD 48 kHz test set relative to published full-band references. Its PESQ, STOI, CBAK, COVL, and SI-SDR values are above the listed PercepNet and DeepFilterNet2 references, while the simplified DeepFilterNet2 reference remains slightly higher in CSIG and the DeepFilterNet2 rows report lower MACs. The available DeepFilterNet2 SI-SDR values are included as official-checkpoint evaluation context. Because most reference-row values are taken from prior papers rather than locally re-evaluated, the table should be read as a published-reference comparison rather than a strict direct re-evaluation.

4.2.2 Wideband Results

We next evaluate the 16 kHz configuration on a standard wideband benchmark to compare enhancement quality and computational cost.

Table 4.2 reports the 16 kHz VoiceBank+DEMAND results. The FastEnhancer-M row is the paper-reported reference. TF-Rehancer (on) denotes the compact FastGRU-based configuration, and TF-Rehancer (off) denotes the non-streaming TimeMHSA-EFN configuration. The on/off labels distinguish model configurations and do not indicate measured real-time or streaming performance.

Method	Params↓	MACs	DNSMOS	DNSMOS(P.835)			SCOREQ↓	SI-SDR↑	PESQ↑	STOI↑	ESTOI↑
			(P.808)↑	SIG↑	BAK↑	OVL↑					
FastEnhancer-M [20]	0.49M	2.90G	3.48	3.39	4.02	3.11	0.243	19.40	3.24	0.950	0.873
TF-Rehancer (on)	0.47M	1.89G	3.49	3.41	4.04	3.13	0.244	19.44	3.33	0.950	0.872
TF-Rehancer (off)	0.62M	2.00G	3.49	3.41	4.06	3.14	0.242	19.72	3.40	0.953	0.876

Table 4.2: VoiceBank+DEMAND 16 kHz test results. FastEnhancer-M uses paper-reported values [20], while TF-Rehancer rows are locally evaluated on the 824-utterance VBD test set. MACs are model-side estimates and exclude STFT/iSTFT.

Relative to the paper-reported FastEnhancer-M reference, TF-Rehancer (on) uses fewer model-side MACs at a similar parameter scale. It reports higher PESQ, SI-SDR, and DNSMOS SIG, BAK, and OVL values, while FastEnhancer-M remains slightly better in SCOREQ and ESTOI. TF-Rehancer (off) reports lower SCOREQ and higher PESQ, SI-SDR, STOI, ESTOI, DNSMOS BAK, and DNSMOS OVL values at higher parameter count and MACs than TF-Rehancer (on). Therefore, the 16 kHz result is consistent with the intended interpretation: the proposed design reports competitive values in a compact configuration and higher quality metrics in a non-streaming TimeMHSA-EFN configuration.

4.2.3 Low-Band Cutoff

We next vary the low-band cutoff from 3 kHz to 8 kHz under the 48 kHz ablation setting. All cutoff variants use the same model size and are evaluated on the VBD 48 kHz test set.

Cutoff	MACs	DNSMOS (P.808)↑	DNSMOS SIG↑	DNSMOS(P.835) BAK↑	DNSMOS(P.835) OVL↑	SI-SDR↑	PESQ↑	STOI↑	ESTOI↑	CSIG↑	CBAK↑	COVL↑	LSD↓	MCD↓
8 kHz	3.81G	3.46	3.39	3.98	3.08	16.50	3.31	0.946	0.866	4.42	3.60	3.90	0.75	2.38
7 kHz	3.54G	3.45	3.39	3.96	3.07	16.11	3.29	0.947	0.867	4.40	3.57	3.88	0.75	2.35
6 kHz	3.28G	3.46	3.39	3.97	3.07	16.75	3.29	0.947	0.868	4.42	3.60	3.89	0.72	2.26
5 kHz	3.02G	3.46	3.38	3.98	3.07	16.83	3.29	0.947	0.868	4.40	3.60	3.88	0.72	2.27
4 kHz	2.76G	3.45	3.38	3.97	3.07	16.29	3.27	0.947	0.867	4.41	3.56	3.87	0.73	2.31
3 kHz	2.50G	3.45	3.40	3.92	3.06	15.41	3.23	0.946	0.867	4.40	3.51	3.84	0.78	2.38

Table 4.3: Low-band cutoff ablation. All rows use the 48 kHz VCTK-DNS setting. Parameter counts are identical and omitted; MACs are model-side estimates.

The cutoff sweep shows a trade-off among computation, perceptual quality, and spectral fidelity. Larger cutoffs increase computation and improve some perceptual scores, whereas smaller cutoffs reduce MACs but can degrade full-band spectral fidelity as measured by LSD and MCD. Under the controlled 48 kHz ablation setting, we use 5 kHz as the default cutoff because it provides a balanced operating point between computational cost, SI-SDR, and full-band spectral metrics.

4.2.4 Architecture Ablations

Table 4.4 compares the default TF-Rehancer architecture with two variants: one without decoder MHCA and one that replaces complex TF filtering with direct spectrum mapping. This ablation is intended to examine the effects of decoder cross-attention and filtering-based output estimation.

Method	DNSMOS	DNSMOS(P.835)			SI-SDR↑	PESQ↑	STOI↑	ESTOI↑	CSIG↑	CBAK↑	COVL↑	LSD↓	MCD↓
	(P.808)↑	SIG↑	BAK↑	OVL↑									
TF-Rehancer	3.45	3.38	3.98	3.07	16.70	3.29	0.948	0.868	4.40	3.59	3.88	0.73	2.28
w/o Decoder MHCA	3.45	3.39	3.96	3.05	16.27	3.14	0.944	0.858	4.35	3.52	3.77	0.77	2.50
w/ Direct Mapping	3.44	3.35	3.96	3.03	14.90	3.05	0.940	0.853	4.24	3.40	3.66	0.78	2.52

Table 4.4: 48 kHz architecture ablation on the VoiceBank+DEMAND 48 kHz test set. All models are trained using VCTK clean speech and DNS full-band noise and evaluated on the VBD 48 kHz test set. The table is used for controlled component analysis rather than external full-band system ranking.

The direct mapping head is weaker than complex TF filtering across the reported metric families. This suggests that filtering-based output estimation is more reliable than direct spectrum mapping in the 48 kHz ablation setting. The w/o decoder MHCA variant remains stronger than direct mapping, but it is lower than TF-Rehancer on most full-reference and spectral metrics. DNSMOS P.808 is tied after rounding, and DNSMOS SIG is slightly higher for the w/o decoder MHCA row. Thus, the controlled ablation shows that decoder cross-attention is more effective than removing it for the main full-reference and spectral trends, while the broader low-band-guided refinement path remains indirectly assessed.

Chapter 5

Conclusion

5.1 Summary

This thesis presented TF-Rehancer, a time-frequency-domain speech enhancement architecture for computationally controlled full-band processing. The main idea is low-band-guided full-band refinement: TF-Rehancer concentrates heavier modeling on low-band speech structure, keeps observed upper-band evidence in the decoder path, updates high-band decoder tokens using low-band encoder context, and estimates the enhanced spectrum through complex time-frequency filtering over the original noisy STFT.

The 48 kHz VoiceBank+DEMAND results show that the DNS4-trained online TF-Rehancer configuration obtains competitive full-reference scores relative to published PercepNet and DeepFilterNet2 references. The 16 kHz VoiceBank+DEMAND benchmark shows a competitive quality-efficiency trade-off. The compact FastGRU-based setting uses fewer model-side MACs than the paper-reported FastEnhancer-M reference at a similar

parameter scale. It reports higher PESQ, SI-SDR, and DNSMOS P.835 values, while FastEnhancer-M remains slightly stronger in SCOREQ and ESTOI. In the controlled 48 kHz ablation setting, direct spectrum mapping degrades full-reference, spectral, and non-intrusive metrics compared with complex TF filtering, suggesting that filtering-based output estimation is more reliable for the proposed architecture. The decoder MHCA ablation shows that decoder cross-attention is more effective than removing it for the main full-reference and spectral trends. The low-band cutoff analysis further shows that 5 kHz provides a balanced operating point between computational cost, waveform fidelity, and full-band spectral fidelity.

Overall, these results support low-band-guided full-band refinement as a computationally controlled design direction for full-band speech enhancement. The architecture preserves observed upper-band information, avoids heavy full-band encoding, and uses full-resolution complex TF filtering to produce the enhanced spectrum.

5.2 Limitations and Future Work

The 48 kHz comparison includes paper-reported full-band references. TF-Rehancer is evaluated locally. A fully aligned protocol with locally re-evaluated baselines remains necessary for stronger external full-band claims. The 48 kHz ablation experiments in this work are designed for controlled component analysis. Future work should therefore evaluate TF-Rehancer and competing full-band systems under a shared training and evaluation protocol, including aligned metric computation and runtime measurement.

Several architectural questions also remain open. The present ablation supports complex TF filtering as the output estimator and characterizes the effect of decoder MHCA.

Further isolating the lightweight high-band adaptation from the subsequent decoder refinement would clarify this component, for example by removing or replacing the local high-band adapter while preserving the observed high-band embedding. In addition, the low-band cutoff was selected as a balanced operating point under the 48 kHz ablation setting; broader datasets, noise conditions, and sampling rates may require adaptive or data-dependent cutoff strategies.

Finally, practical deployment requires further validation beyond model-side MACs. Future work should include real-time runtime profiling, latency analysis, and evaluation on device-oriented scenarios. Extension to multi-channel or reverberant conditions is another possible direction.

These extensions would further clarify the role of low-band-guided full-band refinement in practical speech enhancement.

Bibliography

- [1] H. Liu, X. Liu, Q. Kong, Q. Tian, Y. Zhao, D. Wang, C. Huang, and Y. Wang, “Voicefixer: A unified framework for high-fidelity speech restoration,” in *Proc. Interspeech*, pp. 4232–4236, 2022.
- [2] P. Andreev, A. Alanov, O. Ivanov, and D. Vetrov, “Hifi++: A unified framework for bandwidth extension and speech enhancement,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1–5, 2023.
- [3] J.-M. Valin, U. Isik, N. Phansalkar, R. Giri, K. Helwani, and A. Krishnaswamy, “A perceptually-motivated approach for low-complexity, real-time enhancement of fullband speech,” in *Proc. Interspeech*, pp. 2482–2486, 2020.
- [4] H. Schröter, A. Maier, A. N. Escalante-B., and T. Rosenkranz, “Deepfilternet2: Towards real-time speech enhancement on embedded devices for full-band audio,” in *Proc. International Workshop on Acoustic Signal Enhancement (IWAENC)*, pp. 1–5, 2022.
- [5] H. Erdogan, J. R. Hershey, S. Watanabe, and J. L. Roux, “Phase-sensitive and recognition-boosted speech separation using deep recurrent neural networks,” in

- Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 708–712, 2015.
- [6] D. S. Williamson, Y. Wang, and D. Wang, “Complex ratio masking for monaural speech separation,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 24, no. 3, pp. 483–492, 2016.
- [7] Y. Hu, Y. Liu, S. Lv, M. Xing, S. Zhang, Y. Fu, J. Wu, B. Zhang, and L. Xie, “Dccrn: Deep complex convolution recurrent network for phase-aware speech enhancement,” in *Proc. Interspeech*, pp. 2472–2476, 2020.
- [8] Y. Luo and N. Mesgarani, “Conv-tasnet: Surpassing ideal time-frequency magnitude masking for speech separation,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 27, no. 8, pp. 1256–1266, 2019.
- [9] R. Cao, S. Abdulatif, and B. Yang, “Cmgan: Conformer-based metric gan for speech enhancement,” in *Proc. Interspeech*, pp. 936–940, 2022.
- [10] Y.-X. Lu, Y. Ai, and Z.-H. Ling, “Mp-senet: A speech enhancement model with parallel denoising of magnitude and phase spectra,” in *Proc. Interspeech*, pp. 3834–3838, 2023.
- [11] K. Saijo, G. Wichern, F. G. Germain, Z. Pan, and J. L. Roux, “Tf-locoformer: Transformer with local modeling by convolution for speech separation and enhancement,” in *Proc. International Workshop on Acoustic Signal Enhancement (IWAENC)*, pp. 205–209, 2024.
- [12] P. G. Stelmachowicz, A. L. Pittman, B. M. Hoover, and D. E. Lewis, “Effect of

- stimulus bandwidth on the perception of /s/ in normal- and hearing-impaired children and adults,” *Journal of the Acoustical Society of America*, vol. 110, no. 4, pp. 2183–2190, 2001.
- [13] B. B. Monson, A. J. Lotto, and B. H. Story, “Detection of high-frequency energy level changes in speech and singing,” *Journal of the Acoustical Society of America*, vol. 135, no. 1, pp. 400–406, 2014.
- [14] B. C. J. Moore, “A review of the perceptual effects of hearing loss for frequencies above 3 khz,” *International Journal of Audiology*, vol. 55, no. 12, pp. 707–714, 2016.
- [15] H. Liu, K. Chen, Q. Tian, W. Wang, and M. D. Plumbley, “Audiosr: Versatile audio super-resolution at scale,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1076–1080, 2024.
- [16] Y. Luo and J. Yu, “Music source separation with band-split rnn,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 1893–1901, 2023.
- [17] W.-T. Lu, J.-C. Wang, Q. Kong, and Y.-N. Hung, “Music source separation with band-split rope transformer,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 481–485, 2024.
- [18] X. Rong, T. Sun, X. Zhang, Y. Hu, C. Zhu, and J. Lu, “Gtcrn: A speech enhancement model requiring ultralow computational resources,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 971–975, 2024.
- [19] H. Yan, J. Zhang, C. Fan, Y. Zhou, and P. Liu, “Lisennet: Lightweight sub-band and

- dual-path modeling for real-time speech enhancement,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1–5, 2025.
- [20] S. Ahn, J. Han, B. J. Woo, and N. S. Kim, “Fastenhancer: Speed-optimized streaming neural speech enhancement,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026.
- [21] U.-H. Shin, J. Ko, W. Jeong, and H.-M. Park, “Query-based asymmetric modeling with decoupled input-output rates for speech restoration,” in *Proc. International Conference on Machine Learning (ICML)*, 2026. To appear.
- [22] U.-H. Shin and H.-M. Park, “Asymmetric encoder-decoder based on time-frequency correlation for speech separation,” *arXiv preprint arXiv:2603.29097*, 2026.
- [23] C. Veaux, J. Yamagishi, and K. MacDonald, “Cstr vctk corpus: English multi-speaker corpus for cstr voice cloning toolkit,” 2017.
- [24] C. K. A. Reddy, V. Gopal, R. Cutler, E. Beyrami, R. Cheng, H. Dubey, S. Matusych, R. Aichner, A. Aazami, S. Braun, P. Rana, S. Srinivasan, and J. Gehrke, “The interspeech 2020 deep noise suppression challenge: Datasets, subjective testing framework, and challenge results,” in *Proc. Interspeech*, pp. 2492–2496, 2020.